

## TE – 105 ENGINEERING THERMODYNAMICS

### UNIT III

1. **State the Kelvin – Plank statement of second law thermodynamics. (NOV 2017, MAY 2019, NOV 2016, MAY 2016, JAN 2016)**

It is impossible for a heat engine to produce net work in a complete cycle if it exchanges heat only with bodies at a single fixed temperature.

2. **Define Entropy. (NOV 2017, MAY 2018)**

Entropy, the measure of a system's heat energy per unit temperature that is unavailable for doing useful work. Because work is obtained from ordered molecular motion, the amount of entropy is also a measure of the molecular disorder, or randomness of a system

3. **Why Carnot cycle is a theoretical cycle. (MAY 2019)**

In Carnot cycle the entropy always increases due to the 2nd law of Thermodynamics. In the Carnot cycle, you have isentropic processes and the heat transfer into the engine in the isothermal process is too slow for practical applications.

4. **What is meant by reversible process? (NOV 2016, MAY 2016)**

A reversible process is carried out infinitely slowly with an infinitesimal gradient, so that every state passed through by the system is an equilibrium state. So a reversible process coincides with a quasi - static process.

5. **State Carnot theorem. (JAN 2015)**

It states that of all heat engines operating between a given constant temperature source and a given constant temperature sink, none has a higher efficiency than a reversible engine.

6. **Define Heat pump. (JAN 2015, NOV 2018)**

A heat pump is a device which, operating in a cycle, maintains a body, at a temperature higher than the temperature of the surroundings. By virtue of the temperature difference, there will be heat leakage from the body to the surroundings. The body will be maintained at the constant temperature

7. **Define the term COP? (JAN 2016)**

Coefficient of performance is ratio between desired effects to the work input

$$\text{COP} = \text{Desired effect} / \text{work input (or)}$$

$$\text{COP} = Q_2 / W$$

**8. What is Carnot cycle? What are the processes which constitute the cycle? (NOV 2018, MAY 2017)**

A reversible cycle is an ideal hypothetical cycle in which all the processes constituting the cycle are reversible. Carnot cycle is a reversible cycle. For a stationary system, as in a piston and cylinder machine, the cycle consists of the following four successive processes

- Reversible isothermal expansion process
- Reversible adiabatic expansion process
- Reversible isothermal compression process
- Reversible adiabatic compression process

**9. What is meant by heat engines? (MAY 2017)**

A heat engine is a system that converts heat or thermal energy to mechanical energy, which can then be used to do mechanical work. It does this by bringing a working substance from a higher state temperature to a lower state temperature.

**10. Write Clausius statement for second law of thermodynamics. (MAY 2018)**

It is impossible to construct a device which operating in a cycle will produce no effect other than the transfer of heat from a cooler to a hotter body.

**11. Differentiate between a refrigerator and heat pump. (SEP 2020)**

A refrigerator is a device, which operates in a cycle and maintains the temperature of a particular body lower than the surrounding temperature. And in heat pump, the desired effect is to maintain temperature of the body more than the surrounding temperature

A refrigeration system cools the external fluid flowing through the evaporator, whereas a heat pump heats the external fluid flowing through the condenser

**12. It is possible for a heat engine to operate without rejecting any waste heat to a low-temperature reservoir? Explain (SEP 2020)**

It is not possible for a heat engine to operate without rejecting any waste heat to a low temperature reservoir.

As per Kelvin-Planck statement of the second law of thermodynamics it states that "It is impossible for a heat engine to produce net work in a complete cycle if it exchanges heat only with bodies at a single fixed temperature.

1. A reversible engine is supplied with heat from two constant temperature sources at 900 K and 600 K and rejects heat to a constant temperature sink at 300 K. The engine develops work equivalent to 90 KJ/s and rejects heat at the rate of 56 Kw Estimate :

- (a) Heat supplied by each source, and  
(b) Thermal efficiency of the engine

(NOV2017)

Given

$$T_1 = 900\text{K}$$

$$T_2 = 600\text{K}$$

$$T_3 = 300\text{K}$$

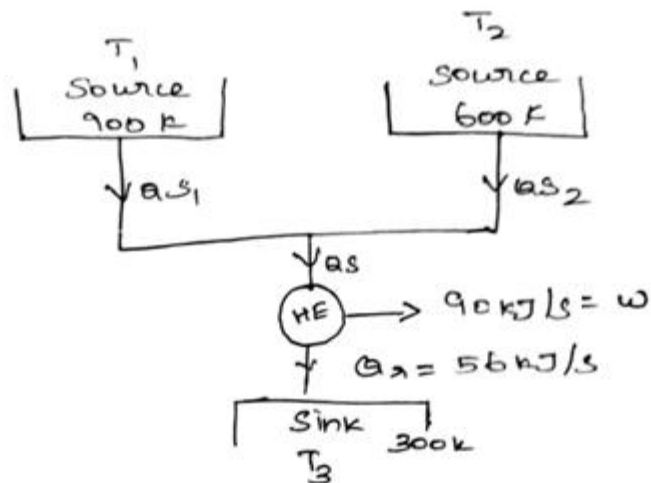
$$Q_R = 56\text{kJ/s}$$

$$W = 90\text{kJ/s}$$

To find

$$Q_{S1}, Q_{S2}, \eta_{HE}$$

Sol



$$\eta_{HE} = \frac{W}{Q_S}$$

$$W = Q_S - Q_R$$

$$90 = Q_S - 56$$

$$Q_S = 146\text{ kJ/s}$$

$$\eta = \frac{W}{Q_S} = \frac{90}{146}$$

$$= 0.616$$

$$= 61.6\%$$

By clausius method

$$\frac{Q_s}{T_1} = \frac{Q_r}{T_2}$$

$$\frac{Q_{s1}}{T_1} + \frac{Q_{s2}}{T_2} = \frac{Q_r}{T_3}$$

$$\frac{Q_{s1}}{900} + \frac{Q_{s2}}{600} = \frac{56}{300}$$

$$\frac{Q_{s1}}{900} + \frac{Q_{s2}}{600} = 0.186 \quad \text{--- (1)}$$

$$Q_s = Q_{s1} + Q_{s2}$$

$$146 = Q_{s1} + Q_{s2}$$

$$Q_{s1} = 146 - Q_{s2} \quad \text{--- (2)}$$

Sub (2) in (1)

$$\frac{146 - Q_{s2}}{900} + \frac{Q_{s2}}{600} = 0.186$$

$$\frac{87,600 - 600Q_{s2} + 900Q_{s2}}{540000} = 0.186$$

$$87,600 - 600Q_{s2} + 900Q_{s2} = 100440$$

$$300Q_{s2} = \frac{100440}{87,600}$$

$$Q_{s2} = \frac{12840}{300}$$

$$\boxed{Q_{s2} = 42.8 \text{ kJ}} \quad \text{--- (3)}$$

Sub (3) in (2)

$$Q_{s1} = 146 - 42.8$$

$$\boxed{Q_{s1} = 103.2}$$

2. A reversible heat engine operates between two reservoirs at  $727^{\circ}\text{C}$  and  $27^{\circ}\text{C}$ . Engine drives a Carnot refrigerator- maintaining at  $-23^{\circ}\text{C}$  and rejecting heat to the reservoir at  $27^{\circ}\text{C}$ . Heat input to the engine is  $2000\text{ kJ}$  and the network available is  $300\text{ kJ}$ . How much heat is transferred to the refrigerator and total heat rejected to the reservoir at  $27^{\circ}\text{C}$ . (MAY 2019)

Given data

$$T_1 = 727^{\circ}\text{C} = 1000\text{ K}$$

$$T_2 = 27^{\circ}\text{C} = 300\text{ K}$$

$$T_3 = 27^{\circ}\text{C} = 300\text{ K}$$

$$T_4 = -23^{\circ}\text{C} = 250\text{ K}$$

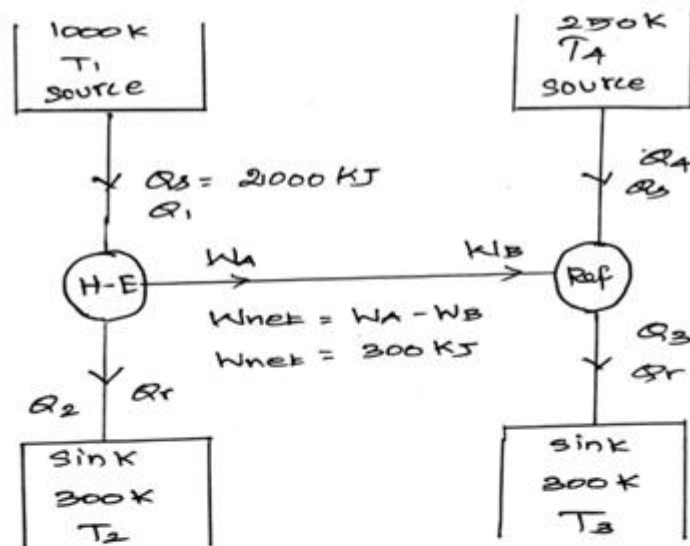
$$Q_3 = 2000\text{ KJ}$$

$$W_{\text{net}} = 300\text{ KJ}$$

To find

$$Q_2, Q_4$$

Sol



$$\eta_A = \frac{W_A}{Q_3} = \frac{Q_3 - Q_2}{Q_3} = \frac{Q_1 - Q_2}{Q_1}$$

$$\eta_A = \frac{T_1 - T_2}{T_1} = \frac{1000 - 300}{1000}$$

$$\eta_A = 70\% \text{ or } 0.7$$

$$\eta_{HE} = 0.7$$

$$COP_{ref} = \frac{Q_S}{W_B} = \frac{Q_A}{Q_3 - Q_A} = \frac{T_A}{T_3 - T_A}$$

$$COP_{ref} = \frac{250}{300 - 250}$$

$$COP_{ref} = 5$$

$$\eta_{HE} = \frac{W_A}{Q_S}$$

$$0.7 = \frac{W_A}{2000}$$

$$W_A = 1400 \text{ KJ}$$

$$W_{HE} = W_A - W_B$$

$$300 = 1400 - W_B$$

$$W_B = 1100 \text{ KJ}$$

$$COP_{ref} = \frac{Q_S}{W_B} = \frac{Q_A}{W_B}$$

$$5 = \frac{Q_A}{1100}$$

$$Q_A = 5500 \text{ KJ} = Q_r$$

$$W_B = Q_r - Q_S = Q_3 - Q_A$$

$$Q_r = W_B + Q_S$$

$$= 1100 + 5500$$

$$Q_3 = Q_r = 6600 \text{ KJ}$$

$$W_A = Q_S - Q_r$$

$$W_A = Q_1 - Q_2$$

$$1400 = 2000 - Q_2$$

$$Q_2 = 600 \text{ KJ} = Q_r$$

Heat transfer to sink at  $27^\circ\text{C}$  (or)  $300\text{K}$

$$Q_2 + Q_3 = 600 + 6600$$

$$Q_2 + Q_3 = 7200 \text{ KJ}$$

3. State briefly the differences between the reversible and irreversible processes.(MAY 2019)

1. Reversible or ideal process
2. Irreversible or natural process

A *reversible* process is one which is performed in such a way that at the conclusion of the process, both the system and the surroundings may be restored to their initial states, without producing any changes in the rest of the universe. Let the state of a system be represented by  $A$  (Fig. 6.12), and let the system be taken to state  $B$  quasistatically by following the path  $A-B$ . If the system and also the surroundings are restored to their initial states on reverse, and no change in the universe is produced, then the process

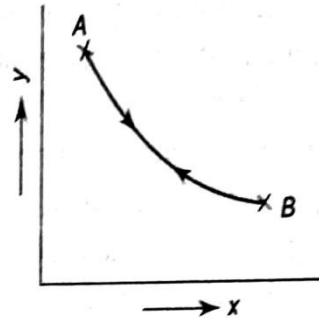


Fig. 6.12 A reversible process

$A-B$  will be a reversible process. In the reverse process, the system has to be taken from state  $B$  to  $A$  by following the same path  $B-A$ . A reversible process should not leave any trace or relic behind to show that the process had ever occurred.

A *reversible process* is carried out infinitely slowly with an infinitesimal gradient, so that every state passed through by the system is an equilibrium state. So a reversible process coincides with a quasi-static process.

Any natural process carried out with a finite gradient is an irreversible process. A reversible process, which consists of a *succession of equilibrium states*, is an idealised hypothetical process, approached only as a limit. It is said to be an *asymptote to reality*. All spontaneous processes are irreversible.

Time has an important effect on reversibility. If the time allowed for a process to occur is infinitely large, even though the gradient is finite, the process becomes reversible. However, if this time is squeezed to a finite value, the finite gradient makes the process irreversible.



4. A reversible heat engine operates between two reservoirs at temperatures  $700^{\circ}\text{C}$  and  $50^{\circ}\text{C}$ . The engine drives a reversible refrigerator which operates between reservoirs at temperatures of  $50^{\circ}\text{C}$  and  $-25^{\circ}\text{C}$ . The heat transfer to the engine is  $2500\text{ kJ}$  and the network output of the combined engine refrigerator plant is  $400\text{ kJ}$ .
- Determine the heat transfer to the • refrigerant and the net heat transfer to the reservoir at  $50^{\circ}\text{C}$ .
  - Reconsider (a) given that the efficiency of the heat engine and the C.O.P. of the refrigerator, are each 45 per cent of their maximum possible values. (NOV 2016)

Given data

$$T_1 = 700^{\circ}\text{C} = 973\text{K}$$

$$T_2 = 50^{\circ}\text{C} = 323\text{K}$$

$$T_3 = 50^{\circ}\text{C} = 323\text{K}$$

$$T_4 = -25^{\circ}\text{C} = 248\text{K}$$

$$Q_1 = Q_3 = 2500\text{ kJ}$$

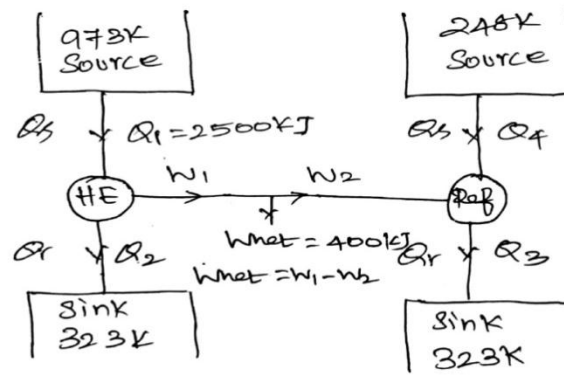
$$W_{\text{net}} = 400\text{ kJ}$$

To find

a)  $Q_2, Q_3, Q_4$

b) Reconsider (a) given that the efficiency of the heat engine and COP of the refrigerator are each 45% of their max. possible values.

Sol



$$\eta_{\text{HE}} = \frac{W_1}{Q_1} = \frac{Q_3 - Q_2}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$

$$\eta_{\text{HE}} = \frac{T_1 - T_2}{T_1} = \frac{973 - 323}{973}$$

$$\eta_{\text{HE}} = 0.668 \quad 66.8\%$$

$$\eta_{\text{HE}} = W_1 / Q_1 = W_1 = \eta_{\text{HE}} \times Q_1$$



$$W_1 = 0.668 \times 2500$$

$$W_1 = 1670 \text{ KJ}$$

$$\text{COP}_{\text{ref}} = \frac{Q_5}{W_2}$$

$$= \frac{Q_5}{Q_1 - Q_5}$$

$$= \frac{Q_4}{Q_3 - Q_4} \cdot \frac{T_4}{T_3 - T_4}$$

$$= \frac{248}{323 - 248}$$

$$\text{COP}_{\text{ref}} = 3.306$$

$$W_{\text{net}} = W_1 - W_2$$

$$400 = 1670 - W_2$$

$$W_2 = 1270 \text{ KJ}$$

$$\text{COP}_{\text{ref}} = Q_4 / W_2$$

$$3.306 = Q_4 / 1270$$

$$Q_4 = 4099.44 \text{ KJ}$$

$$W_2 = Q_3 - Q_4$$

$$1270 = Q_3 - 4099.44$$

$$Q_3 = 5369.44 \text{ KJ}$$

$$W_1 = Q_1 - Q_2$$

$$1670 = 2500 - Q_2$$

$$Q_2 = 830 \text{ KJ}$$

Heat rejection to the  $45^\circ\text{C}$  reservoir.

$$Q_2 + Q_3 = 830 + 5369.44$$

$$Q_2 + Q_3 = 6199.44 \text{ KJ}$$

Energy balance for refrigerator.

Input Energy = output Energy

$$W_2 + Q_4 = Q_3$$

$$W_2 = Q_3 - Q_4$$

b) Efficiency of the actual heat engine cycle.

$$\eta_{HE \text{ act}} = 0.45 \times \eta_{\max}$$

$$= 0.45 \times 0.668$$

$$\eta_{\text{act}} = 0.3006$$

$$\eta_{\text{act}} = W_1 / Q_3$$

$$0.3006 = W_1 / 2500$$

$$W_1 = \underline{751.5 \text{ kJ}}$$

$$W_{\text{net}} = W_1 - W_2$$

$$400 = 751.5 - W_2$$

$$W_2 = \underline{351.5 \text{ kJ}}$$

$$W_1 = Q_1 - Q_2$$

$$751.5 = 2500 - Q_2$$

$$Q_2 = \underline{1748.5 \text{ kJ}}$$

COP<sub>act</sub> efficiency

$$COP_{\text{act}} = \frac{Q_4}{W_2}$$

$$1.487 = \frac{Q_4}{351.5}$$

$$Q_4 = \underline{522.68 \text{ kJ}}$$

$$COP_{\text{act}} = 0.45 \times 3.306$$

$$= 1.487$$

$$W_2 = Q_3 - Q_4$$

$$351.5 = Q_3 - 522.68$$

$$Q_3 = \underline{874.18 \text{ kJ}}$$

Heat rejected to the 40°C reservoir

$$Q_2 + Q_3 = 1748.5 + 874.18$$

$$\boxed{Q_2 + Q_3 = 2622.68 \text{ kJ}}$$

5. A reversible heat engine operates between two reservoirs at temperatures of  $600^{\circ}\text{C}$  and  $40^{\circ}\text{C}$ . The engine drives a reversible refrigerator which operates between reservoirs at temperatures of  $40^{\circ}\text{C}$  and  $-20^{\circ}\text{C}$ . The heat transfer to the heat engine is  $2000\text{ kJ}$  and the net work output of the combined engine refrigerator plant is  $360\text{ kJ}$ .
- (a) Evaluate the heat transfer to the refrigerant and the net heat transfer to the reservoir at  $40^{\circ}\text{C}$ .
- (b) Reconsider (a) given that the efficiency of the heat engine and the COP of the refrigerator are each  $40\%$  of their maximum possible values. (MAY 2016)

Given data

$$T_1 = 600^{\circ}\text{C} = 873\text{ K}$$

$$T_2 = 40^{\circ}\text{C} = 313\text{ K}$$

$$T_3 = 40^{\circ}\text{C} = 313\text{ K}$$

$$T_4 = -20^{\circ}\text{C} = 253\text{ K}$$

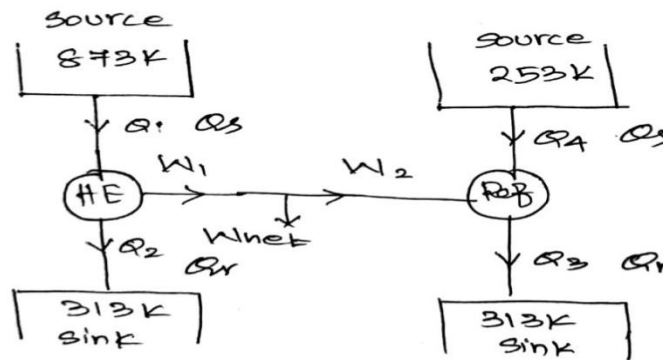
$$Q_1 \text{ or } Q_3 = 2000\text{ kJ}$$

$$W_{\text{net}} = 360\text{ kJ}$$

To Find

- a)  $Q_2, Q_3, Q_2 + Q_3$ ,  
b) Reconsider given efficiency of heat engine and COP are each  $40\%$  of their maximum values.

Sol



$$\eta_{\text{HE}} = \frac{W_1}{Q_3} = \frac{Q_1 - Q_2}{Q_1} = \frac{T_1 - T_2}{T_1} = \frac{T_H - T_L}{T_H}$$

$$= \frac{873 - 313}{873}$$

$$\eta_{\text{HE}} = 0.642 \quad 64.2\%$$

$$\eta_{\text{HE}} = \frac{W_1}{Q_3}$$

$$W_1 = 0.642 \times 2000$$

$$\underline{W_1 = 1284 \text{ KJ}}$$

$$\text{COP}_{\text{ref}} = \frac{Q_4}{W_2} = \frac{Q_4}{Q_3 - Q_4}$$

$$= \frac{T_4}{T_3 - T_4}$$

$$= \frac{253}{313 - 253}$$

$$\underline{\text{COP}_{\text{ref}} = 4.22}$$

Energy balance for  
refrigerator  
input Energy = output  
Energy

$$W_2 + Q_4 = Q_3$$

$$W_2 = Q_3 - Q_4$$

$$W_{\text{net}} = W_1 - W_2$$

$$360 = 1284 - W_2$$

$$\underline{W_2 = 924 \text{ KJ}}$$

$$\text{COP}_{\text{ref}} = \frac{Q_4}{W_2}$$

$$4.22 = Q_4 / 924$$

$$\underline{Q_4 = 3899 \text{ KJ}}$$

$$W_2 = Q_3 - Q_4$$

$$924 = Q_3 - 3899$$

$$\underline{Q_3 = 4823 \text{ KJ}}$$

$$W_1 = Q_1 - Q_2$$

$$1284 = 2000 - Q_2$$

$$\underline{Q_2 = 716 \text{ KJ}}$$

Heat rejection to the  $40^\circ\text{C}$  reservoir.

$$Q_2 + Q_3 = 716 + 4823$$

$$\boxed{Q_2 + Q_3 = 5539 \text{ KJ}}$$

b) Efficiency of the actual Heat Engine cycle.

$$\eta_{HE} = 0.4 \eta_{max}$$

$$= 0.4 \times 0.642$$

$$\eta_{act} = 0.256$$

$$\eta_{act} = W_1 / Q_1$$

$$0.256 = W_1 / 2000$$

$$W_1 = 512 \text{ KJ}$$

$$W_{net} = W_1 - W_2$$

$$360 = 512 - W_2$$

$$W_2 = 152 \text{ KJ}$$

$$W_1 = Q_1 - Q_2$$

$$512 = 2000 - Q_2$$

$$\boxed{Q_2 = 1488 \text{ KJ}}$$

COP of the actual efficiency.

$$COP_{act} = \frac{Q_4}{W_2}$$

$$COP_{act} = 0.4 \times 4.22$$

$$0.4 \times 4.22 = \frac{Q_4}{152}$$

$$1.69 = \frac{Q_4}{152}$$

$$\boxed{Q_4 = 259.6 \text{ KJ}}$$

$$W_2 = Q_3 - Q_4$$

$$153.6 = Q_3 - 259.6$$

$$\boxed{Q_3 = 413.2 \text{ KJ}}$$

Heat rejected to the  $40^\circ\text{C}$  reservoir

$$Q_2 + Q_3 = 1488 + 413.2$$

$$\boxed{Q_2 + Q_3 = 1899.6 \text{ KJ}}$$

## 6. Discuss the inequality of Clausius and give its expression for reversible and irreversible processes. (MAY 2016, NOV 2018, MAY 2018)

### 7.6 THE INEQUALITY OF CLAUSIUS

Let us consider a cycle  $ABCD$  (Fig. 7.10). Let  $AB$  be a general process, either reversible or irreversible, while the other processes in the cycle are reversible. Let the cycle be divided into a number of elementary cycles, as shown. For one of these elementary cycles

$$\eta = 1 - \frac{dQ_2}{dQ}$$

where  $dQ$  is the heat supplied at  $T$ , and  $dQ_2$  the heat rejected at  $T_2$ .

Now, the efficiency of a general cycle will be equal to or less than the efficiency of a reversible cycle.

$$\therefore 1 - \frac{dQ_2}{dQ} \leq \left( 1 - \frac{dQ_2}{dQ} \right)_{\text{rev}}$$

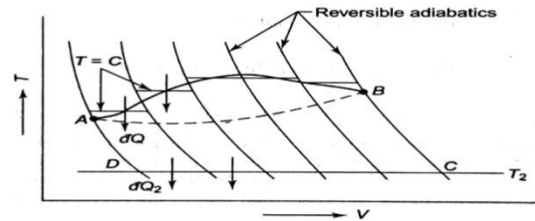


Fig. 7.10 Inequality of Clausius

$$\text{or } \frac{dQ_2}{dQ} \geq \left( \frac{dQ_2}{dQ} \right)_{\text{rev}}$$

$$\text{or } \frac{dQ}{dQ_2} \leq \left( \frac{dQ}{dQ_2} \right)_{\text{rev}}$$

$$\text{Since } \left( \frac{dQ}{dQ_2} \right)_{\text{rev}} = \frac{T}{T_2}$$

$$\therefore \frac{dQ}{dQ_2} \leq \frac{T}{T_2}$$

or  $\frac{dQ}{T} \leq \frac{dQ_2}{T_2}$ , for any process  $AB$ , reversible or irreversible.

For a reversible process

$$ds = \frac{dQ_{\text{rev}}}{T} = \frac{dQ_2}{T_2} \quad (7.7)$$

Hence, for any process  $AB$

$$\frac{dQ}{T} \leq ds \quad (7.8)$$

Then for any cycle

$$\oint \frac{dQ}{T} \leq \oint ds$$

Since entropy is a property and the cyclic integral of any property is zero

$$\oint \frac{dQ}{T} \leq 0 \quad (7.9)$$

This equation is known as the *inequality of Clausius*. It provides the criterion of the reversibility of a cycle.

If  $\oint \frac{dQ}{T} = 0$ , the cycle is reversible,

$\oint \frac{dQ}{T} < 0$ , the cycle is irreversible and possible

$\oint \frac{dQ}{T} > 0$ , the cycle is impossible, since it violates the second law.

[*Aliter*: Let us consider two heat engine cycles, one reversible and the other irreversible, operating between the same two constant temperature reservoirs  $T_1$  and  $T_2$ . For the supply of the same quantity of heat  $Q_1$  to both the engines, and since the efficiency of the irreversible engine  $\eta_I$  is less than that of the reversible engine  $\eta_R$

$$\eta_I < \eta_R$$

$$\frac{W_I}{Q_1} < \frac{W_R}{Q_1}$$

$$W_I < W_R$$

$$\therefore Q_1 - Q_2' < Q_1 - Q_2$$

$$\therefore Q_2' > Q_2$$

where  $Q_2$  is the heat rejection from the reversible engine and  $Q_2'$  is that from the irreversible engine. For the reversible engine,

$$\oint \frac{dQ}{T} = \frac{Q_1}{T_1} - \frac{Q_2}{T_2} = 0$$

For the irreversible engine,

$$\oint \frac{dQ}{T} = \frac{Q_1}{T_1} - \frac{Q_2'}{T_2} < 0$$

Similarly, it can be proved that  $\oint \frac{dQ}{T} < 0$  for any cyclic device operating between more than two reservoirs.

$$\text{So, in general } \oint \frac{dQ}{T} \leq 0$$

The equality sign holds good for any reversible cycle and the inequality is true for an irreversible cycle. The integration has to be performed over the entire cycle. If for a cycle,  $\oint \frac{dQ}{T} > 0$ , that cycle is impossible. The Clausius inequality is also valid for refrigerators and heat pumps.]

7. A reversible engine operates between a source at  $972^\circ\text{C}$  and two sinks one at  $127^\circ\text{C}$  and another at  $27^\circ\text{C}$ . The energy rejected is same at the both sinks. What is the ratio of heat supplied to the heat rejected? Also calculate the efficiency. (JAN 2015)

Given data

$$T_1 = 972^\circ\text{C} = 1245\text{K}$$

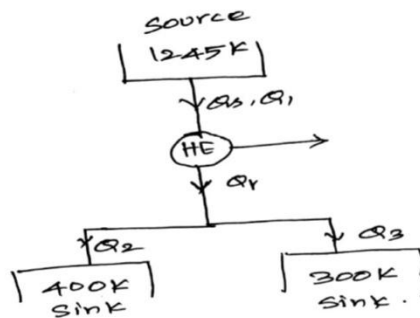
$$T_2 = 127^\circ\text{C} = 400\text{K}$$

$$T_3 = 27^\circ\text{C} = 300\text{K}$$

$$Q_2 = Q_3$$

$$\text{To find } Q_1/Q_2 \text{ or } \frac{Q_1}{Q_2} = ? , ?$$

So:



$$\frac{Q_1}{T_1} - \frac{Q_2}{T_2} - \frac{Q_3}{T_3} = 0$$

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2} + \frac{Q_3}{T_3} \quad Q_2 = Q_3$$

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2} + \frac{Q_2}{T_3}$$

$$\frac{Q_1}{1245} = \frac{Q_2}{400} + \frac{Q_2}{300}$$

$$\frac{Q_1}{1245} = \frac{706 Q_2}{120000}$$

$$\frac{Q_1}{1245} = \frac{7 Q_2}{1200}$$

$$\frac{Q_1}{Q_2} = \frac{7 \times 1245}{1200} = 7.265$$



$$\frac{Q_1}{Q_2} = 7.265$$

$$\eta_{HE} = ?$$

$$\eta = \frac{W}{Q_s} = \frac{W}{Q_1} = \frac{Q_1 - 2Q_2}{Q_1}$$

$$W = Q_s - 2Q_r$$

$$Q_1/Q_2 = 7.265$$

$$Q_2 = \frac{Q_1}{7.265}$$

$$W = Q_1 - 2Q_2$$

$$= Q_1 - 2 \frac{Q_1}{7.265}$$

$$= Q_1 - \frac{2Q_1}{7.265}$$

$$= \frac{7.265Q_1 - 2Q_1}{7.265}$$

$$= \frac{5.265Q_1}{7.265}$$

$$W = 0.724 Q_1$$

$$\eta = \frac{W}{Q_1} = \frac{0.724 Q_1}{Q_1}$$

$$\boxed{\eta = 0.72 \quad 72\%}$$

8. An irreversible heat engine with 66% efficiency of the maximum possible is operating between 1000K and 300K. If it delivers 3 kW of work, determine the heat extracted from the high temperature reservoir and heat rejected to low temperature reservoir. (JAN 2016)

Given data

$$\eta = 66\% \quad 0.66$$

$$T_L = 300 \text{ K}$$

$$T_H = 1000 \text{ K}$$

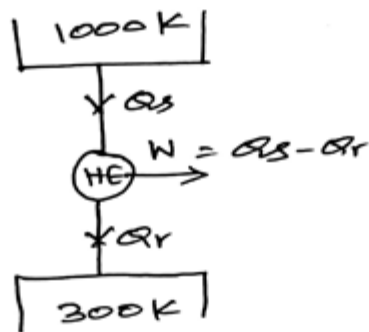
$$W = 3 \text{ kW}$$

To Find

$Q_S$  (heat Extracted).

$Q_R$  (heat rejected).

Sol



$$\eta_{HE} = 1 - \frac{T_L}{T_H}$$

$$= 1 - \frac{300}{1000}$$

$$\eta_{HE} = 0.7$$

$$\eta_{act} = 0.7 \times 0.66$$

$$\eta_{act} = 0.462$$

$$\eta_{act} = W/Q_S = 0.462 = \frac{3}{Q_S}$$

$$\boxed{Q_S = 6.493 \text{ kW}}$$

$$W = Q_S - Q_R$$

$$3 = 6.493 - Q_R$$

$$\boxed{Q_R = 3.493 \text{ kW}}$$

9. A Carnot refrigerator is working between reservoirs of  $-30^{\circ}\text{C}$  and  $44^{\circ}\text{C}$ . what is the co-efficient of performance? If the actual refrigerator has a co-efficient of performance that is 70 percent of the Carnot value, estimate the refrigerant effect per kW of power consumption. (MAY 2017)

Given data

$$T_1 = -30^{\circ}\text{C} = 243\text{K}$$

$$T_2 = 44^{\circ}\text{C} = 317\text{K}$$

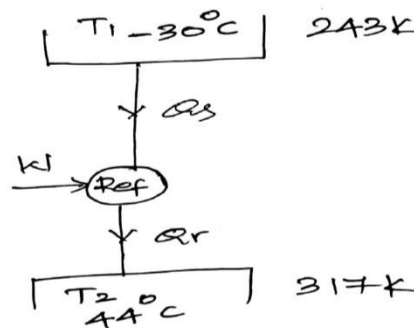
$$\text{COP}_{\text{actual}} = 70\% = 0.7$$

$$W = 1\text{KW}$$

To find

Refrigeration effect per kW

Sol



$$\text{COP}_{\text{ref}} = \frac{Q_s}{W} = \frac{Q_s}{Q_r - Q_s} = \frac{T_1}{T_2 - T_1}$$

$$\text{COP}_{\text{ref}} = \frac{243}{317 - 243}$$

$$\text{COP}_{\text{ref}} = 3.283$$

$$\text{COP}_{\text{act}} = 3.283 \times 0.7$$

$$\text{COP}_{\text{act}} = 2.298$$

$$\text{COP}_{\text{act}} = \frac{\text{Refrigeration effect}}{\text{Work input}}$$

$$2.298 = \frac{R.E}{1}$$

$$\text{Refrigeration effect (R.E)} = 2.298\text{KW.}$$

10. Determine whether the following cases represent the reversible, irreversible or impossible heat engines, (a) 800 kW of heat rejected (b) 570 kW of heat rejected (c) 150 kW of heat rejected. In each case the engine is supplied with 1140 kJ/s of heat. The source and sink temperature are maintained at 740K and 370K respectively.(SEP 2020)

Given data

$$Q_s \text{ (or) } Q_1 = 1140 \text{ kW}$$

$$T_1 = 740 \text{ K}$$

$$T_2 = 370 \text{ K}$$

To find

Following case represent the reversible  
Irreversible or Impossible Heat Engine.

a)  $Q_r = 800 \text{ kW}$

b)  $Q_r = 570 \text{ kW}$

c)  $Q_r = 150 \text{ kW}$ .

Sol

a) 800 kW of heat rejected

By clausius method.

$$\frac{Q_s}{T_1} = \frac{Q_r}{T_2}$$

$$\frac{1140}{740} = \frac{800}{370}$$

$$1.54 = 2.162$$

$$1.54 - 2.162 = 0$$

$$= -0.622$$

$$\frac{dQ}{T} < 0$$

The Engine are Irreversible

b) 570 kW of heat rejected

$$\frac{Q_s}{T_1} = \frac{Q_r}{T_2}$$

$$\frac{1140}{740} = \frac{570}{370}$$

$$1.54 = 1.54$$

$$1.54 - 1.54 = 0$$

$$\frac{dq}{T} = 0$$

The Engines are reversible

c) 150 kW of heat rejected.

$$\frac{Q_s}{T_1} = \frac{Q_r}{T_2}$$

$$\frac{1140}{740} = \frac{150}{370}$$

$$1.54 = 0.405$$

$$1.54 - 0.405 = 0$$

$$1.135$$

$$\frac{dq}{T} > 0$$

The Engines are Impossible.

11. An engine is supplied with 1120 kJ/s of heat. The source and sink temperature are maintained at 560K and 280K. Determine whether the following cases represent the reversible, irreversible or impossible heat engines, (a) 900 kW of heat rejected (b) 560 kW of heat rejected (c) 108 kW of heat rejected. (JAN 2016)

Given data

$$Q_1 = Q_2 = 1120 \text{ kJ/s}$$

$$T_1 = 560 \text{ K}$$

$$T_2 = 280 \text{ K}$$

To find

determine following cases represents  
reversible, Irreversible and Impossible.

- So! a) 900 kW of heat rejected  
b) 560 kW of heat rejected  
c) 108 kW of heat rejected.

sol

- a) 900 kW of heat rejected

By Clausius method

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

$$\frac{1120}{560} = \frac{900}{280}$$

$$2 = 3.21$$

$$2 - 3.21 = 0$$

$$-1.21$$

$$\frac{dQ}{T} < 0$$

The Engine are Irreversible.

- b) 560 kW of heat rejected

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

$$\frac{1120}{560} = \frac{560}{280}$$

$$2 = 2$$

$$2 - 2 = 0$$

$dQ/T = 0$  the Engine are reversible.

- c) 108 kW of heat rejected

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

$$\frac{1120}{560} = \frac{108}{280}$$

$$2 = 0.385$$

$$2 - 0.385 = 0$$

$$1.615$$

$$dQ/T > 0$$

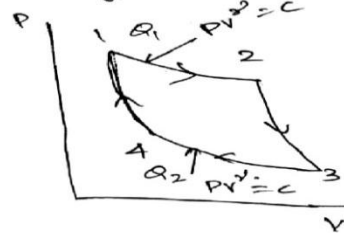
The Engine are Impossible

12. Prove that the efficiency of the Carnot cycle is  $[(T_1 - T_2)/T_1]$ , where  $T_1 > T_2$ . Also draw the p-v and T-s diagram of Carnot cycle. (NOV 2017)

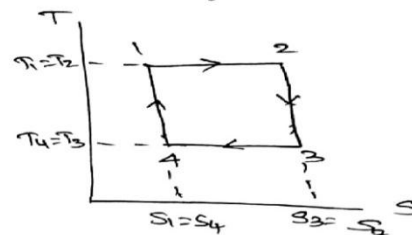
A Reversible cycle is an Ideal hypothetical cycle in which all the processes constituting the cycle are Reversible. For a stationary system, as in a piston & cylinder machine, the cycle consists of the following process.

- 1-2 Reversible Isothermal Expansion
- 2-3 Reversible Adiabatic Expansion
- 3-4 Reversible Isothermal Compression
- 4-1 Reversible adiabatic Compression.

Pv diagram



Ts diagram.



$$\eta_{\text{Carnot}} = \frac{Q_{1-2} - Q_{3-4}}{Q_{1-2}} = 1 - \frac{Q_{3-4}}{Q_{1-2}}$$

for process 1-2 R.I.T.P

$$Q_{1-2} = W_{1-2} = mRT_1 \ln \frac{V_2}{V_1}$$

$$Q_{3-4} = -W_{3-4} = -mRT_2 \ln \frac{V_4}{V_3} \quad T_2 = T_3$$

for 3-4 process R.I.T.C.P

$$\eta = 1 - \frac{mRT_2 \ln(V_3/V_4)}{mRT_1 \ln(V_2/V_1)}$$

4-1 process

$$\frac{T_4}{T_1} = \left(\frac{V_1}{V_4}\right)^{\gamma-1} \quad \text{--- (1)}$$



2-3 process

$$\frac{T_3}{T_2} = \left(\frac{V_2}{V_3}\right)^{\gamma-1} \quad \text{--- (2)}$$

Equ. (1) &amp; (2)

$$\left(\frac{V_1}{V_4}\right)^{\gamma-1} = \left(\frac{V_2}{V_3}\right)^{\gamma-1} \quad T_2 = T_3$$

$$T_1 = T_4$$

$$\left(\frac{V_1}{V_2}\right)^{\gamma-1} = \left(\frac{V_4}{V_3}\right)^{\gamma-1}$$

$$\eta = 1 - \frac{m/R T_2 \ln\left(\frac{V_2}{V_1}\right)}{m/R T_1 \ln\left(\frac{V_2}{V_1}\right)^{\gamma-1}}$$

$$\eta_{\text{carnot}} = 1 - \frac{T_2}{T_1} = \frac{T_1 - T_2}{T_1}$$

$$\eta_{\text{carnot}} = 1 - \frac{T_L}{T_H}$$

$$T_2 < T_1$$

13. An inventor claims that his proposed engine will develop 40kW for a heat addition of 60kW. The source and sink temperatures are maintained at 1800K and 600K respectively. Would you agree this claim. (SEP 2020)

Given data

$$W = 40 \text{ kW}$$

$$Q_S = 60 \text{ kW}$$

$$T_H = 1800 \text{ K}$$

$$T_L = 600 \text{ K}$$

To find

Inventor valid or not.

Sol

$$\eta_{HE(\text{inventor})} = W/Q_S$$

$$\eta_{\text{inventor}} = 40/60$$

$$\eta_{\text{inv}} = 0.666 \quad 66.6\%$$

$$\eta_{\text{carnot}} = 1 - \frac{T_L}{T_H}$$

$$= 1 - \frac{600}{1800}$$

$$\eta_{\text{carno}} = 0.666 \quad 66.6\%$$

$$\eta_{\text{carnot}} = \eta_{\text{given}}$$

Reversible process.

Inventor claim is valid.

14. Find the change in entropy of steam generated at 400°C from 5 kg of water at 27°C and atmospheric pressure. Take specific heat of water to be 4.2 kJ/kg.K, heat of vaporization at 100°C as 2260 kJ/kg and Specific heat for steam given by  $C_p = r(3.5 + 1.27T + 0.14T^2)$  J/kg.K. (NOV 2016) (Out of syllabus)
15. Two kg of water at 80°C are mixed adiabatically with 3 kg of water at 30°C in a constant pressure of 1 atmosphere. Find the increase in the entropy of the total mass of water due to the mixing process ( $C_p$  of water = 4.187 KJ/kg.K). (NOV 2018) (Out of syllabus)
16. A vessel contains superheated steam at 6 MPa and 500°C. The vessel is cooled until the pressure of steam drops to 10 bar. Determine the entropy change and heat transfer for the process per unit mass of steam (MAY 2017) (Refer unit 2 Q. No. 22)
17. A domestic food refrigerator maintains a temperature of  $-12^\circ\text{C}$ . The ambient air temperature is  $35^\circ\text{C}$ . If heat leaks into the freezer at the continuous rate of 2 kJ/s determine the least power necessary to pump this heat out continuously. (JAN 2015)

Solution. Freezer temperature,

$$T_2 = -12 + 273 = 261 \text{ K}$$

Ambient air temperature,

$$T_1 = 35 + 273 = 308 \text{ K}$$

Rate of heat leakage into the freezer = 2 kJ/s

Least power required to pump the heat :

The refrigerator cycle removes heat from the freezer at the same rate at which heat leaks into it (Fig. 5.12).

For minimum power requirement

$$\frac{Q_2}{T_2} = \frac{Q_1}{T_1}$$

$$\therefore Q_1 = \frac{Q_2}{T_2} \times T_1 = \frac{2}{261} \times 308 = 2.36 \text{ kJ/s}$$

$$\therefore W = Q_1 - Q_2 = 2.36 - 2 = 0.36 \text{ kJ/s} = 0.36 \text{ kW}$$

Hence, least power required to pump the heat continuously  
= 0.36 kW. (Ans.)

